bargaining problem is still given by (8), and hence $\pi(x) = y(x) - w(x)$, where $y(x) \equiv [\max(x - \phi, 0) - c]k$ is output net of the variable cost and the rental on capital.

The values of unemployment and employment to a worker are

$$rU = b + \theta q(\theta) \iint \max[W(z) - U, 0] dG(z \mid x) dH(x) \tag{33}$$

$$rW(x) = w(x) + \lambda \int \max[W(z) - U, 0] dG(z \mid x) - (\delta + \lambda)[W(x) - U], \tag{34}$$

where $w(x)$ is still characterized by (9). Letting $S(x) = J(x) + W(x) - U$ denote the surplus from a match, notice that (9) implies $J(x) = (1-\beta)S(x)$, $W(x) - U = \beta S(x)$, $\pi(x) = (1 - \beta)[y(x) - rU]$ and

$$w(x) = \beta y(x) + (1-\beta) rU. \tag{35}$$

Combining (9) with (32), (33), and (34) yields

$$(r + \delta + \lambda) S(x) = y(x) - rU + \lambda \int \max[S(z), 0] dG(z \mid x),$$

where rU is given by (10). The fact that $S'(x) > 0$ implies that there exists a unique R such that $S(x) > 0$ iff $x > R$. Hence matches separate (or do not form) for productivity draws below R. Moreover, notice that $S(x)$ is strictly increasing for all x (even for $x < \phi$) despite the fact that $y(x)$ is flat for $x < \phi$. This is because $dG(\cdot \mid x)$ is stochastically increasing in x. Consequently, the equilibrium may have $R > \phi$ (no hoarding) for some parametrizations and $R < \phi$ (hoarding) for others.[41]

Next, I turn to the issue of aggregation. Suppose idiosyncratic shocks are draws from

$$G(x \mid s) = \begin{cases} 0 & \text{if } x < \varepsilon(s) \\ 1 - \left[\frac{\varepsilon(s)}{x}\right]^{\alpha} & \text{if } \varepsilon(s) \le x, \end{cases}$$

where $\varepsilon(\cdot)$ is a continuously differentiable function and $\alpha > 1$. I introduce positively correlated shocks by assuming that $\varepsilon' > 0$ (if $\varepsilon' = 0$ the distribution reduces to (20), the special case of uncorrelated shocks). In addition, suppose there is an $\underline{\varepsilon} > 0$ such that $\varepsilon(\underline{\varepsilon}) = \underline{\varepsilon}$, and $\varepsilon(s) = 0$ if $s < \underline{\varepsilon}$ and that $\lim_{s \to \infty} \varepsilon(s) = 1 + \underline{\varepsilon} \equiv \overline{\varepsilon}$.[42]

Then for $R \ge \varepsilon(s)$, $1 - G(R \mid s) = \left[\frac{\varepsilon(s)}{R}\right]^{\alpha}$; hence, for any $x \ge R$, $G(x \mid s) - G(R \mid s) = [\varepsilon(s)/R]^{\alpha}[1 - (R/x)^{\alpha}]$. After substituting these expressions in (30), it becomes clear that the steady-state productivity distribution of active matches is still given by (21). So for this case, the job-creation and job-destruction conditions are

$$\frac{\mu^{1-\alpha} R^{\alpha}}{\alpha - 1} - \frac{\alpha[y(R) - rU]}{k}\left[\frac{R}{\varepsilon(R)}\right]^{\alpha} \int_R \frac{\varepsilon(x)^{\alpha} - \varepsilon(R)^{\alpha}}{x^{1+\alpha}} dx - \frac{(r+\delta+\lambda)c}{(1-\beta)q(\theta)}\left[\alpha \int_R \frac{\varepsilon(x)^{\alpha}}{x^{1+\alpha}} dx\right]^{-1} = 0$$

$$\left[1 - \frac{\lambda\alpha}{r+\delta+\lambda} \int_R \frac{\varepsilon(x)^{\alpha} - \varepsilon(R)^{\alpha}}{x^{1+\alpha}} dx\right] \frac{y(R) - rU}{k} + \frac{\lambda}{r+\delta+\lambda} \frac{\varepsilon(R)^{\alpha} \mu^{1-\alpha}}{\alpha - 1} = 0.$$

41. In trying to determine which parametrizations may exhibit hoarding, the logic of Figure 3 in Appendix B also applies to this case.

42. An example of a function $\varepsilon(\cdot)$ satisfying all these conditions is $\varepsilon(s) = 1 + \underline{\varepsilon} - e^{\underline{\varepsilon} - s}$ for any $\underline{\varepsilon} > 0$.

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An equilibrium is still a list $[R, \theta, H, U, w, u, K]$ such that R, θ, and H jointly solve (30) and the job-creation and job-destruction conditions, rU is given by (10), w is given by (35), and u satisfies (29). In addition, the market for capital should clear, so $[1-(1-\theta)u]k = K$. Sufficient conditions can be found so that the job-creation condition slopes downward and the job-destruction condition slopes upward in θ–R space, implying a unique (θ, R) pair. A parameter restriction analogous to the one depicted in Figure 3 in Appendix B, which guarantees that ϕ has a range of values such that $R < \phi$ can still be derived.[43] Given the equilibrium pair (θ, R), the same procedure followed in Section 4 reveals that output aggregates to (22). Thus, the key aggregation result obtained in the context of the model of Section 2 does not hinge on some of the particular modelling choices made there. In particular, allowing for correlated shocks can generate hoarding, and there is no need for the fixed cost of Section 2.

8.2. *More on aggregation*

In this subsection I prove two additional aggregation results. First, I provide another specification of primitives that delivers the same aggregation result obtained for the models of Sections 2 and 8.1. Again, this specification does not rely on the fixed cost of Section 2, so in a way it reinforces the message of Section 8.1. The second aggregation result shows how to reverse engineer a distribution of idiosyncratic shocks that gives rise to an aggregate constant elasticity of substitution (CES) production function.

First consider the model of Section 2, but with $C = 0$ and separation rates that are decreasing in the productivity of the match; that is, $0 < \delta(x) < \infty$ for all x, with $\delta' < 0$. The interpretation is that δ is a technological parameter: in any small time interval, with probability $\delta(x)$ the idiosyncratic productivity of the match jumps to zero and stays at that level forever. I return to the case of uncorrelated productivity draws for the remainder of the section. The distribution of active matches now evolves according to

$$\begin{aligned}\frac{d}{dt}[(1-u_t)H_t(x)] = {} & \lambda(1-u_t)[1-H_t(x)][G(x)-G(R_t)] + \theta q(\theta)u_t[G(x)-G(R_t)] \\ & -\lambda(1-u_t)H_t(x)G(R_t) - \lambda(1-u_t)H_t(x)[1-G(x)] \\ & -(1-u_t)\int_R \delta(s)dH_t(s). \end{aligned} \tag{36}$$

The value functions (5) and (7) are respectively replaced by

$$rW(x) = w(x) + \lambda\int \max[W(z)-U, 0]dG(z) - [\delta(x)+\lambda][W(x)-U]$$

$$rJ(x) = \pi(x) + \lambda\int \max[J(z)-V, 0]dG(z) - [\delta(x)+\lambda][J(x)-V],$$

while (4) and (6) remain unchanged. The bargaining outcome is still characterized by (8) and (9), and the value functions imply

$$S(x) = \frac{[\max(x-\phi, 0)-c]k - rU + \lambda\int \max[S(z), 0]dG(z)}{r+\delta(x)+\lambda},$$

43. Showing that equilibria with $R < \phi$ are possible for some parametrizations is now rather tedious, so the basic idea is only outlined here. Let $\phi_{\overline{\varepsilon}}$ be the value of ϕ such that $\theta^*_{\overline{\varepsilon}}$ and $R(\phi_{\overline{\varepsilon}}) = \overline{\varepsilon}$ solve the job-creation and job-destruction conditions. Then if $\phi_{\overline{\varepsilon}} - \overline{\varepsilon} > 0$, there will be an interval $(\phi_{\overline{\varepsilon}}, \widehat{\phi})$ such that $R(\phi) < \phi$ iff $\phi \in (\phi_{\overline{\varepsilon}}, \widehat{\phi})$. If, in addition, $\partial R(\phi)/\partial\phi > 0$, then $\phi_{\overline{\varepsilon}} - \overline{\varepsilon} > 0$ also implies $\overline{\varepsilon} < R(\phi)$ for all ϕ. Finally, notice that $R > \overline{\varepsilon}$ also implies that *every* match faces a positive probability of being destroyed for endogenous reasons. To see why, suppose $R = \varpi < \overline{\varepsilon}$; then any match that reaches a state $s > \varepsilon^{-1}(\varpi)$ will never be destroyed endogenously.

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which is clearly increasing in x and has a kink at ϕ, as is usual when departing from the simple fixed-cost formulation of Section 2. Plotting $S(x)$ reveals that, depending on parametrizations, two cases are possible: the kink could be above or below the horizontal axis. If it is above, then $R < \phi$ and the equilibrium exhibits hoarding. The job-creation and job-destruction conditions can be derived as usual, and an equilibrium can be summarized by the (θ, R) pair that solves them. The following result provides conditions under which this model aggregates to (22), just as in the models of Sections 2 and 6.1.

Proposition 3. *Suppose $\delta(x) = \delta x^{-\sigma}$ and the primitive density of shocks is*

$$dG(x) = \frac{\alpha\lambda(\alpha+\sigma)\varepsilon^{\alpha}}{\lambda(\alpha+\sigma)+\alpha\delta\varepsilon^{-\sigma}}\left(1+\frac{\delta}{\lambda}x^{-\sigma}\right)x^{-\alpha-1}$$

for $x \geq \varepsilon$ and $dG(x) = 0$ otherwise, where $\varepsilon, \delta, \sigma > 0$, and $\alpha > 1$. Then in equilibrium, the aggregates Y, K_e, and N satisfy (22).

So far I have shown that in several versions of the Mortensen–Pissarides model, for certain distributions of the idiosyncratic shocks, aggregate output looks like a Cobb–Douglas function of the aggregate labour and capital inputs. What follows generalizes the previous results by characterizing the distribution of shocks that gives rise to an aggregate CES production function.

Suppose the primitive distribution of shocks, G, is given by

$$G(x) = \begin{cases} 0 & \text{if } x < \varepsilon \\ 1-\left[\frac{1}{\sigma}\left(\frac{x}{\varepsilon}\right)^{\rho/(1-\rho)} - \frac{1-\sigma}{\sigma}\right]^{-1/\rho} & \text{if } \varepsilon \leq x, \end{cases} \tag{37}$$

with $\varepsilon > 0$ and $\rho, \sigma \in (0, 1)$.[44] Substituting (37) into (3), one sees that for any $R \geq \varepsilon$, the steady-state productivity distribution of active matches is

$$H(x) = 1-\kappa\left[\frac{1}{\sigma}\left(\frac{x}{\varepsilon}\right)^{\frac{\rho}{1-\rho}} - \frac{1-\sigma}{\sigma}\right]^{-1/\rho} \tag{38}$$

if $R \leq x$, and $H(x) = 0$ if $x < R$, with $\kappa \equiv [1-G(R)]^{-1}$. Following a strategy similar to that of Levhari (1968), one can establish the following result.

Proposition 4. *If the primitive distribution of the idiosyncratic shocks is given by (37), then in equilibrium, the aggregates Y, K_e, and N satisfy $Y = B\left[\sigma\bar{A}K_e^{\rho} + (1-\sigma)N^{\rho}\right]^{1/\rho}$, with $B = \frac{\varepsilon}{1-\sigma}$, and $\bar{A} = \left[\frac{1}{\sigma}\left(\frac{R}{\varepsilon}\right)^{\rho/(1-\rho)} - \frac{1-\sigma}{\sigma}\right]$.*

In this case, all the characteristics of the labour market as summarized by R affect the measured productivity of inputs asymmetrically.[45] Notice that as $\rho \to 0$, (37) approaches the Pareto distribution in (20) with parameters ε and $\alpha = 1/\sigma$. So in this sense, the CES aggregate in Proposition 4 approaches the Cobb–Douglas aggregate in (22) as the elasticity of substitution $1/(1-\rho)$ approaches unity.[46]

44. Under these conditions, $G'(x) \geq 0$ and $\lim_{x\to\infty} G(x) = 1$, so G is a proper cumulative distribution function.

45. If the aggregation were performed using (37) instead of its truncation, then the aggregate would instead be $Y = \frac{R}{1-\sigma}[\sigma K_e^{\rho} + (1-\sigma)N^{\rho}]^{1/\rho}$. However, no primitive density has (37) as its truncation.

46. Notice, however, that the truncation of (37) does not approach (21) as $\rho \to 0$. That is, even though the primitive distribution approaches a Pareto, its truncation does not limit a truncated Pareto. This is because the density in (37) is not "closed" under truncations (as, for example, the Pareto and the exponential distributions are). This "discontinuity" introduced by the truncation is the reason why if we take the limit on the truncated cumulative distribution function or on the CES aggregate directly, we do not obtain exactly (22).

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9. CONCLUDING REMARKS

This paper developed a theory of TFP differences based on the interaction between institutions and the microeconomics underlying the aggregate production function. The analysis focused on a precise class of institutions, namely, labour-market policies as measured by the magnitudes of hiring subsidies, employment subsidies, unemployment benefits, and firing taxes. In the model, firm-level technologies are subject to idiosyncratic shocks that induce a cross-sectional distribution of productivities. Through their effects on job-creation and job-destruction decisions, labour-market policies affect the productivity composition of active firms—and ultimately TFP.

By thinking of the aggregate production function as an equilibrium relationship between aggregate inputs and output arising from the aggregation of heterogeneous micro-level production units, I was able to show analytically how the measured level of TFP—usually a "black box" that drives cross-country income differences—depends on primitive technological parameters, government policies, and other fundamental features of the economic environment. This approach may prove useful in interpreting aggregate productivity data and can serve as a guide to uncovering sources of cross-country differences in measured TFP.

APPENDIX A

Proof of Proposition 1.

In R–θ space, the slopes of the job-destruction and job-creation conditions (24) and (25) are

$$\frac{1-\beta}{\beta c}\left[1-\frac{\lambda(\varepsilon/R)^{\alpha}}{r+\delta+\lambda}\right] \quad \text{and} \quad \frac{(1-\beta)\theta q(\theta)(\varepsilon/R)^{\alpha}\tau(R)}{-c\eta(\theta)(r+\delta+\lambda)R},$$

respectively, where $\tau(R)=R+\alpha(r+\delta+\lambda)(\tau_h-\tau_f)$. Note that for all $R\geq\varepsilon$, the destruction condition slopes upward, and the creation condition slopes downward. If $\phi=\phi_\epsilon$, then (24) and (25) have a unique solution, namely, $\theta(\phi_\epsilon)=\theta_\epsilon^*$ and $R(\phi_\epsilon)=\varepsilon$. Increases in ϕ only shift the destruction condition downward, increasing the equilibrium level of R and decreasing the equilibrium level of θ (the creation condition is independent of ϕ). In addition, for any given ϕ, the creation condition (25) asymptotes the horizontal axis in (R,θ) space and the job-destruction condition (24) grows without bound. Therefore, (a)–(d) follow for any $\phi>\phi_\epsilon$. Finally, $\phi_\epsilon-\varepsilon>0$ is equivalent to $\phi_\epsilon-R(\phi_\epsilon)>0$, which implies (e). ||

Proof of Proposition 2. Define $\Delta=\frac{(\varepsilon/R)^{\alpha}\tau(R)}{(r+\delta+\lambda)R}+\frac{\eta(\theta)}{\beta\theta q(\theta)}\left[1-\frac{\lambda(\varepsilon/R)^{\alpha}}{r+\delta+\lambda}\right]$. Since $\tau(R)>0$ by Proposition 1, it follows that $\Delta>0$ in any equilibrium. By totally differentiating (24) and (25),

$$\frac{\partial R}{\partial\tau_e}=\frac{-\eta(\theta)}{\beta\theta q(\theta)\Delta}<0,\ \frac{\partial R}{\partial\tau_f}=-(1/\Delta)\left[(\varepsilon/R)^{\alpha}+\frac{r\eta(\theta)}{\beta\theta q(\theta)}\right]<0,$$
$$\frac{\partial R}{\partial\tau_h}=(1/\Delta)(\varepsilon/R)^{\alpha}>0,\ \frac{\partial R}{\partial\tau_b}=-\frac{\partial R}{\partial\tau_e}>0,$$

and this concludes the proof. ||

Proof of Proposition 3. First, note that the corresponding distribution function is

$$G(x)=\frac{\alpha\lambda(\alpha+\sigma)\varepsilon^{\alpha}}{\lambda(\alpha+\sigma)+\alpha\delta\varepsilon^{-\sigma}}\left\{\frac{\varepsilon^{-\alpha}}{\alpha}\left[1-\left(\frac{\varepsilon}{x}\right)^{\alpha}\right]+\frac{\delta\varepsilon^{-(\alpha+\sigma)}}{\lambda(\alpha+\sigma)}\left[1-\left(\frac{\varepsilon}{x}\right)^{\alpha+\sigma}\right]\right\}. \tag{39}$$

Verify that $G(\varepsilon)=0$ and $\lim_{x\to\infty}G(x)=1$. Under the parametric restrictions in the statement, $dG(x)\geq 0$ for all x, so dG is a proper density. The restriction $\alpha>1$ ensures the mean is finite. Imposing steady states in (36), substituting (39) and solving for $H(x)$ reveals that $H(x)$ is as in (21). ||

Proof of Proposition 4. The problem is to find a cumulative distribution function H that satisfies $H(R)=0$ and yields

$$Y=a\left[\sigma_1(\hat{\kappa}K_e)^{\rho}+\sigma_2N^{\rho}\right]^{1/\rho}, \tag{40}$$

where $\rho\in(0,1)$ and a, $\hat{\kappa}$, σ_1, and σ_2 are positive constants. Define $\varsigma(x)=\int_x zh(z)dz$ and $s(x)=1-H(x)$. Since, in general, $Y=\varsigma(\mu)K_e$ and $N=s(\mu)K_e$, (40) can be rewritten as $\varsigma(x)^{\rho}=a^{\rho}\left[\sigma_1\hat{\kappa}^{\rho}+\sigma_2s(x)^{\rho}\right]$. Differentiating both sides of

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